

Guideline on doxycycline post-exposure prophylaxis for sexually transmitted infections among men who have sex with men and transgender women



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Abbreviations

CI	confidence interval
PEP	post-exposure prophylaxis
PrEP	pre-exposure prophylaxis
RR	relative risk
STI	sexually transmitted infection

Executive summary

Sexually transmitted infections (STIs) remain a major global public health challenge, with limited progress in reducing incidence despite decades of effort. New evidence-informed prevention tools are urgently needed. Doxycycline post-exposure prophylaxis (PEP) – a single 200-mg dose taken within 72 hours after condomless sex – has emerged as a promising strategy to prevent bacterial STIs. This guideline presents WHO's first recommendation on doxycycline PEP for STI prevention among men who have sex with men and transgender women, who experience disproportionately high STI burdens.

Based on review of the available evidence by the Guideline Development Group, WHO issues a conditional recommendation that doxycycline PEP may be offered to men who have sex with men and transgender women as part of a comprehensive sexual health package complementing other STI prevention strategies. Importantly, given the concerns about how prophylactic antibiotic use affects antimicrobial resistance, WHO strongly encourages that context-appropriate monitoring of antimicrobial resistance be performed alongside doxycycline PEP policy implementation.

1. Introduction

Sexually transmitted infections (STIs) continue to impose a substantial global public health burden. Despite sustained efforts over several decades, progress in reducing STI incidence has remained limited, underscoring the ongoing need for new evidence-informed tools.

Prevention is a cornerstone of public health, but options for preventing STIs remain limited. Using medicines to prevent infections (prophylaxis) is a well-established strategy for infectious diseases, including HIV. Doxycycline, which is active against bacterial STIs, may offer an important preventive approach when used as prophylaxis. Doxycycline post-exposure prophylaxis (PEP) involves taking a single dose of 200 mg of doxycycline within 72 hours of condomless sex to prevent bacterial STIs. This guideline provides a recommendation on the use of doxycycline PEP for STI prevention among men who have sex with men and transgender women, who are disproportionately affected by STIs. Guidance for other populations and potential use as pre-exposure prophylaxis (PrEP) may be developed in the future as more evidence becomes available.

This guideline summarizes the recommendation, implementation considerations and research gaps related to doxycycline PEP for STI prevention. Its purpose is to strengthen evidence-informed STI prevention among men who have sex with men and transgender women. Full details of the guideline development process, evidence review (with the evidence-to-decision table), formulation of the recommendation, literature list and management of conflicts of interest are available in the Web Annex.

2. Recommendation

Doxycycline PEP may be offered to men who have sex with men and transgender women for prevention of bacterial STIs

(Conditional recommendation, moderate-certainty evidence)



Remarks

- Doxycycline PEP should be provided as part of a comprehensive sexual health approach, complementing other prevention strategies such as condom use.
- Men who have sex with men and transgender women with recent or recurrent STIs – especially syphilis – and those in high-incidence sexual networks or settings should be given priority.
- Context-appropriate monitoring of antimicrobial resistance is strongly encouraged as part of doxycycline PEP policy. Ideally, trends in both antimicrobial resistance and antibiotic use should be monitored.

3. Summary of supporting evidence



Three randomized controlled trials were identified through a quantitative systematic review with data on efficacy and safety (see Web Annex) (1-4). Pooled analysis showed that for men who have sex with men and transgender women, doxycycline PEP reduces the incidence of syphilis (relative risk (RR) 0.20; 95% confidence interval (CI) 0.11–0.37, high certainty) and chlamydia (RR 0.15; 95% CI 0.10–0.23, high certainty) compared with no doxycycline PEP. It also probably reduces the incidence of gonorrhoea (RR 0.52; 95% CI 0.40–0.68; moderate certainty) and any STI defined as either chlamydia, gonorrhoea or syphilis (RR 0.27; 95% CI 0.15–0.47; moderate certainty), although the effect on gonorrhoea depends on the prevalence of circulating tetracycline resistance to *Neisseria gonorrhoeae* within the population (5-6). Doxycycline PEP has similar effects on delaying the time to first infection.

Grade 3 or 4 adverse events probably do not differ between doxycycline PEP and no doxycycline PEP (RR 0.84; 95% CI 0.43–1.67, moderate certainty) (1-4). The effects of doxycycline PEP on antimicrobial resistance are uncertain based on the available data (three randomized controlled trials; four non-randomized studies) (1-4, 7-9); there may be no effect on tetracycline resistance to *Neisseria gonorrhoeae* (RR 1.00; 95% CI 0.94–1.05; low certainty), and no evidence of resistance to cephalosporins or other antibiotic classes was identified. No data were available for several of the important outcomes, and for others – including sexual risk behaviour and antimicrobial resistance outcomes in other bacteria – the effects of doxycycline PEP were uncertain.

A qualitative evidence synthesis of seven studies identified through the systematic review (see Web Annex) provides moderate-certainty evidence that men who have sex with men and transgender women place high value on STI prevention and generally consider their personal risks of antimicrobial resistance and long-term safety acceptable compared with the benefits of doxycycline PEP (10-16). They appreciate its role in reducing anxiety and stigma and supporting safer sex, although concerns about side-effects and microbiome changes persist. Awareness of and willingness to use doxycycline PEP are high, with cost and access identified as key barriers. Acceptability and feasibility are high among men who have sex with men, transgender women and health-care providers, with only limited hesitation among users because of information gaps. These findings were consistent across high- and low-income settings as confirmed by extensive virtual community consultation about values and preferences among men who have sex with men and transgender women WHO conducted across all regions. No studies with cost or cost-effectiveness data were identified.

4. Justification

Balance of effects

High-certainty evidence indicates that doxycycline PEP reduces the incidence of syphilis and chlamydia, and moderate-certainty evidence indicates that it likely reduces the incidence of gonorrhoea and any STI but with smaller effect size. Global differences in tetracycline resistance to *N. gonorrhoeae* suggest that the preventive effect of doxycycline PEP for this infection will be setting specific and depend on local resistance prevalence. The Guideline Development Group considered the strong preventive effect against syphilis the most valuable given the pathogenesis of the infection, gaps in clinical care and its impact on health.

Moderate-certainty evidence indicates no difference in grade 3 or 4 adverse events between doxycycline PEP users and non-users. The evidence available to the Guideline Development Group on doxycycline PEP's effects on antimicrobial resistance in STIs and other pathogens points towards selection for tetracycline resistance to *N. gonorrhoeae* and to other pathogens and changes to the microbiome. However, the extent of this selection and its public health impact remain uncertain, especially because of inadequate data and differences in circulating prevalence of tetracycline resistance to *N. gonorrhoeae* in different populations. Existing studies are few, vary in design, often include a limited range of pathogens and have small sample sizes.

Overall, the Guideline Development Group viewed the balance of desirable and undesirable effects to favour the benefits of doxycycline PEP for STI prevention – combined with no difference in grade 3 or 4 adverse events – over the uncertain impact on antimicrobial resistance.

Values and preferences

Men who have sex with men and transgender women place high value on preventing STIs, reducing anxiety and empowerment, with limited concerns about potential stigma associated with doxycycline PEP use. They acknowledge potential side-effects and generally view long-term antimicrobial resistance risks as acceptable compared with the direct preventive benefits of doxycycline PEP. Awareness of and willingness to use doxycycline PEP are high, and overall, there is probably no important uncertainty or variability in how men who have sex with men and transgender women value these outcomes. Access barriers in low- and middle-income countries (such as prescription rules, pharmacy regulations and stigma) could shape acceptability differently, but the core values remain consistent. WHO community consultation conducted across all regions supported these findings from the qualitative evidence synthesis.

Resources required

The resources required for doxycycline PEP are uncertain and likely to vary across settings. Although the drug cost of doxycycline is generally low, and it is often available in primary health care, additional expenses may arise for implementation and capacity building. Integrating doxycycline PEP into sexual health services could incur further costs, especially in low-resource settings, where scaling up comprehensive sexual health service provision may increase overall expenditure. Long-term costs related to antimicrobial resistance are unknown. Conversely, cost savings may occur through reduced health-care utilization for symptomatic STIs.

Health equity and human rights

Men who have sex with men and transgender women reported willingness to pay for doxycycline PEP but strongly prefer low-cost or free-of-charge access to avoid relying on unregulated sources. Health-care providers emphasized

the importance of ensuring equitable access, acceptability and appropriate use of doxycycline PEP. Clear formal guidance would facilitate availability and probably increase health equity and promote human rights by strengthening trust and service provision between health-care systems, providers and communities.

Acceptability and feasibility

Doxycycline PEP is highly acceptable among users and within communities of men who have sex with men and transgender women, generally acceptable to service providers and public health practitioners (although concerns about antimicrobial resistance remain) and only marginally acceptable to microbiologists. Perspectives may vary across stakeholder groups and globally, but doxycycline PEP is probably acceptable to most stakeholders and in most settings worldwide. Implementation is also considered probably feasible in most settings, given its high efficacy, low cost and overall acceptability among key stakeholders. The importance of educational materials and development of implementation models – including out-of-facility services – to further enhance acceptability and feasibility is emphasized.

Conclusions

Based on the available evidence, the Guideline Development Group conditionally recommended that doxycycline PEP may be offered to men who have sex with men and transgender women for preventing bacterial STIs. Doxycycline PEP should be provided as part of a comprehensive sexual health approach to ensure access to high-quality services. men who have sex with men and transgender women with recent STIs, especially syphilis, and those in high-incidence sexual networks and settings should be given priority. Given the uncertainty around how doxycycline PEP affects antimicrobial resistance – with indications that it increases resistance to tetracycline-class antibiotics with uncertainty around the extent of selection and the impact on other antibiotic classes in STIs, other pathogens and the microbiome – context-appropriate antimicrobial resistance monitoring is strongly encouraged alongside doxycycline PEP policy implementation.

5. Implementation considerations



Timing of doxycycline use

- Doxycycline should be taken as PEP ideally within 24 hours, and no later than 72 hours, after condomless sex to maximize effectiveness. This timing interval is based on trial protocols and reflects doxycycline's pharmacological profile, including differences in drug distribution across different mucosal tissues.
- Use is event-driven – based on exposure – and repeat 200-mg dosing within any 24-hour period should be avoided to reduce the risk of adverse events given the drug's long half-life. Depending on the frequency of use, some people may effectively transition along a continuum between doxycycline PEP and PrEP. There is currently no evidence to support daily use, and WHO does not currently recommend doxycycline for PrEP, but future guidance may be developed as more evidence becomes available.
- Continued monitoring of adverse events and communicating approaches to mitigating side-effects (such as taking doxycycline with food to reduce gastrointestinal discomfort) are important. Nutrient interactions should be considered: products containing iron, magnesium or calcium – including supplements and dairy foods – can reduce doxycycline absorption and should be taken at least a few hours apart from doxycycline.

Setting priorities based on STI risk

- Men who have sex with men and transgender women and men who have sex with men with recent STIs – especially syphilis – and those in high-incidence sexual networks or settings should be given priority because the preventive effect of doxycycline PEP is most powerful for these populations. High-incidence networks

or settings are environments or groups of people in which STIs are circulating at higher than usual levels, increasing the likelihood of exposure because more partners, or partners' partners, are infected.

- Eligibility should be periodically reassessed, considering changes in STI incidence, sexual behaviour, individual preferences and medication tolerance, to support person-centred care and antimicrobial stewardship.
- Cost-effectiveness improves and unnecessary antibiotic use is limited when people with a higher incidence of infection are targeted for doxycycline PEP. Broad, general population-level roll-out of doxycycline PEP without assessing individual STI risk behaviour should be avoided.

Integration into comprehensive health services

- Doxycycline PEP should be integrated into a comprehensive prevention and sexual health approach rather than being offered as a stand-alone intervention. It can complement – but does not replace – other prevention strategies such as condoms, STI testing and partner services.
- Messaging for demand creation should focus on pleasure and empowerment and be non-stigmatizing. Lessons from HIV PrEP implementation highlight the importance of community engagement, peer-led education and outreach in this process.
- Educational materials are important for informed decision-making. Other key elements for high-quality integrated service provision include access to counselling, non-judgemental STI prevention, testing and treatment and linkage to HIV prevention and treatment services as appropriate.

Service delivery models for equitable access

- Locally appropriate service pathways for doxycycline PEP are essential for equitable access, especially if these can ensure low-cost access. These could include primary health care facilities, sexual health clinics and community-based services. Provision through pharmacies (with and without prescription) and online platforms as guided by local laws and regulations may provide additional options.
- Task-sharing of prescription, provision and/or dispensing to lay providers and clarification of prescription terms are important to ensure scalability in low-resource and diverse settings.

Monitoring and evaluation

- Doxycycline PEP strategies should include a robust, context-appropriate approach to monitoring and evaluation. Components should ideally cover user demographics, access pathways, patterns of use and STI incidence indicators.
- Given the concerns about antimicrobial resistance, monitoring for antimicrobial resistance is encouraged alongside doxycycline PEP policy. This may include tracking drug resistance in STIs and to other pathogens and potentially changes in the microbiome. Concurrent monitoring of antibiotic use trends is essential to provide context for any observed change in antimicrobial resistance rates. Doxycycline PEP implementation should be aligned with national antimicrobial resistance strategies and surveillance platforms, especially in settings with limited laboratory capacity.

6. Research gaps

Antimicrobial resistance

The overall impact of doxycycline PEP on antimicrobial resistance is still uncertain because of inadequate data. Although existing studies are few, vary in design and often have small sample sizes, data point towards selection for resistance to tetracycline-class antibiotics. Research priorities include identifying indicator pathogen(s) and drug and/or genomic target(s) for monitoring antimicrobial resistance trends to inform policy evaluation and pre-specifying thresholds for intervention. Other important gaps include studies on resistance emergence in bystander populations and potential cross-resistance to other antibiotics. Importantly, effects should be assessed at the levels of users, the target population and the general population.

Impact on global STI burden

The preventive effects of doxycycline PEP for STIs are well documented among users. Early programme implementation data show similar declines in population-based STI rates in high-resource settings. However, documenting the impact of doxycycline PEP programmes in diverse contexts – including low-resource settings – and across populations, including the general population, remains an important research gap. Further, cost and cost-effectiveness evaluations should be conducted since there is currently no evidence on the resources required.

Drug interactions and dosing strategy

The systematic review found no data on potential interactions with antiretroviral therapy for HIV infection, hormone therapy and opioid agonist treatment – issues of concern to the affected communities. Although such interactions have not been reported when doxycycline is used for STI treatment, they remain undocumented for prolonged prophylactic use. Further, research is needed to determine optimal dosing frequency along the continuum of doxycycline PrEP and PEP for people with frequent sexual exposure (such as sex workers).

Optimizing integrated service delivery models

Models for integrated service delivery should be developed for diverse settings. Opportunities include co-packaging doxycycline PEP with other STI and HIV prevention services and integrating doxycycline PEP into pharmacy-based access points, digital health platforms and app-based services to support remote counselling and subscription. Further, operational guidance in settings in which diagnostic testing is not available should be developed. Research to optimize the delivery modes for doxycycline PEP is a priority.

Implementation research

Doxycycline PEP should be provided through locally acceptable and culturally appropriate service delivery models. Implementation research offers a framework to assess the fidelity, acceptability and scalability of doxycycline PEP programmes in real-world settings and is essential to refine service models and ensure equitable access. Implementation research should include low- and middle-income settings (including task-sharing, pharmacy-based delivery and community-led models), given the diverse systems in which doxycycline PEP may be implemented. Key considerations include cost-effectiveness, sustainability and health equity outcomes alongside preventive effectiveness.

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